

Package ‘india’

September 14, 2026

Type Package

Title Influence Diagnostics in Statistical Models

Version 0.1-5

Date 2026-09-14

Maintainer Felipe Osorio <faosorios.stat@gmail.com>

Description Set of routines for influence diagnostics by using case-deletion in ordinary least squares, nonlinear regression [Ross (1987). <doi:10.2307/3315198>], ridge estimation [Walker and Birch (1988). <doi:10.1080/00401706.1988.10488370>] and least absolute deviations (LAD) regression [Sun and Wei (2004). <doi:10.1016/j.spl.2003.08.018>].

Depends R(>= 3.5.0), fastmatrix, L1pack

LinkingTo fastmatrix

Imports stats

License GPL-3

URL <https://github.com/faosorios/india>

NeedsCompilation yes

LazyLoad yes

Author Felipe Osorio [aut, cre] (ORCID:
<<https://orcid.org/0000-0002-4675-5201>>)

Repository CRAN

Date/Publication 2026-09-14 02:50:17 UTC

Contents

aircraft	2
cooks.distance	2
envelope	4
leverages	6
logLik.displacement	9
portland	11

relative.condition	12
rquantile	13
skeena	14

Index	16
--------------	-----------

aircraft	<i>Aircraft data</i>
----------	----------------------

Description

This dataset is presented in Rousseeuw and Leroy (1987, pp. 154), and the aim is to model the cost of 23 single-engine aircraft (in unit of \$100,000) as a function of the following explanatory variables: aspect ratio, lift-to-drag ratio, weight of the plane, in pounds, and maximal thrust.

Usage

```
data(aircraft)
```

Format

A data frame with 23 observations on the following 5 variables.

- aspect** aspect ratio.
- lift2drag** lift-to-drag ratio.
- weight** weight of the plane (in pounds).
- thrust** maximal thrust.
- cost** cost of 23 single-engine aircraft (in unit of \$100,000).

Source

Rousseeuw, P.J., Leroy, A.M. (1987). *Robust Regression and Outlier Detection*. Wiley, New York.

cooks.distance	<i>Cook's distances</i>
----------------	-------------------------

Description

Cook's distance is a measure to assess the influence of the *i*th observation on the model parameter estimates. This function computes the Cook's distance based on leave-one-out cases deletion for ordinary least squares, nonlinear least squares, lad and ridge regression.

Usage

```
## S3 method for class 'ols'
cooks.distance(model, ...)
## S3 method for class 'nls'
cooks.distance(model, ...)
## S3 method for class 'lad'
cooks.distance(model, type = "L2", ...)
## S3 method for class 'ridge'
cooks.distance(model, type = "cov", ...)
```

Arguments

model	an R object, returned by <code>ols</code> , <code>nls</code> , <code>lad</code> or <code>ridge</code> .
type	For 'lad' objects, specifies the norm used for the influence measure. "L2" computes the Cook's distance based on the Euclidean norm, whereas "L1" computes the absolute-norm fitted-value measure FD_{1i} proposed in Equation (4) of Sun and Wei (2004). For 'ridge' objects, the available options are "1st", "cov" and "both", corresponding to the Cook's distances given by Equations (2.5), (2.6), or both, respectively, in Walker and Birch (1988).
...	further arguments passed to or from other methods.

Value

A vector whose i th element contains the Cook's distance,

$$D_i(\mathbf{M}, c) = \frac{(\hat{\beta}_{(i)} - \hat{\beta})^T \mathbf{M} (\hat{\beta}_{(i)} - \hat{\beta})}{c},$$

for $i = 1, \dots, n$, with $\mathbf{M}$ a positive definite matrix and $c > 0$. Specific choices of $\mathbf{M}$ and c are done for objects of class `ols`, `nls`, `lad` and `ridge`.

The Cook's distance for nonlinear regression is based on linear approximation, which may be inappropriate for expectation surfaces markedly nonplanar.

For `lad` objects with `type = "L1"`, the returned measure is

$$FD_i = \frac{1}{\hat{\omega}} \|\mathbf{X}\hat{\beta} - \mathbf{X}\hat{\beta}_{(i)}\|_1 = \frac{1}{\hat{\omega}} \sum_{j=1}^n |x_j^T \hat{\beta} - x_j^T \hat{\beta}_{(i)}|,$$

where

$$\hat{\omega} = \frac{1}{n-p} \sum_{j=1}^n |y_j - x_j^T \hat{\beta}|.$$

References

- Cook, R.D., Weisberg, S. (1980). Characterizations of an empirical influence function for detecting influential cases in regression. *Technometrics* **22**, 495-508. doi:[10.1080/00401706.1980.10486199](https://doi.org/10.1080/00401706.1980.10486199)
- Cook, R.D., Weisberg, S. (1982). *Residuals and Influence in Regression*. Chapman and Hall, London.

- Ross, W.H. (1987). The geometry of case deletion and the assessment of influence in nonlinear regression. *The Canadian Journal of Statistics* **15**, 91-103. doi:10.2307/3315198
- Sun, R.B., Wei, B.C. (2004). On influence assessment for LAD regression. *Statistics & Probability Letters* **67**, 97-110. doi:10.1016/j.spl.2003.08.018
- Walker, E., Birch, J.B. (1988). Influence measures in ridge regression. *Technometrics* **30**, 221-227. doi:10.1080/00401706.1988.10488370

Examples

```
# Cook's distances for linear regression
fm <- ols(stack.loss ~ ., data = stackloss)
CD <- cooks.distance(fm)
plot(CD, ylab = "Cook's distances", ylim = c(0,0.8))
text(21, CD[21], label = as.character(21), pos = 3)

# Cook's distances for LAD regression
fm <- lad(stack.loss ~ ., data = stackloss)
CD <- cooks.distance(fm, type = "L2")
plot(CD, ylab = "Cook's distances", ylim = c(0,0.4))
text(17, CD[17], label = as.character(17), pos = 3)

# Fitted-value measure, L1-norm for LAD regression
FD <- cooks.distance(fm, type = "L1")
plot(FD, ylab = "Fitted-value distances", ylim = c(0,2.3))
text(17, FD[17], label = as.character(17), pos = 3)

# Cook's distances for ridge regression
data(portland)
fm <- ridge(y ~ ., data = portland)
CD <- cooks.distance(fm)
plot(CD, ylab = "Cook's distances", ylim = c(0,0.5))
text(8, CD[8], label = as.character(8), pos = 3)

# Cook's distances for nonlinear regression
data(skeena)
model <- recruits ~ b1 * spawners * exp(-b2 * spawners)
fm <- nls(model, data = skeena, start = list(b1 = 3, b2 = 0))
CD <- cooks.distance(fm)
plot(CD, ylab = "Cook's distances", ylim = c(0,0.35))
obs <- c(5, 6, 9, 19, 25)
text(obs, CD[obs], label = as.character(obs), pos = 3)
```

envelope

QQ-plot of residuals with simulated envelope

Description

Constructs a normal QQ-plot with simulated envelope of residuals from a fitted model object.

Usage

```

envelope(object, ...)
## S3 method for class 'lm'
envelope(object, reps = 50, conf = 0.95,
  type = c("quantile", "standard", "student"), plot.it = TRUE, ...)
## S3 method for class 'lad'
envelope(object, reps = 50, conf = 0.95, plot.it = TRUE, ...)
## S3 method for class 'ols'
envelope(object, reps = 50, conf = 0.95,
  type = c("quantile", "standard", "student"), plot.it = TRUE, ...)
## S3 method for class 'nls'
envelope(object, reps = 50, conf = 0.95, plot.it = TRUE, ...)
## S3 method for class 'ridge'
envelope(object, reps = 50, conf = 0.95, plot.it = TRUE, ...)

```

Arguments

<code>object</code>	an R object, returned by <code>lm</code> , <code>lad</code> , <code>ols</code> , <code>nls</code> or <code>ridge</code> .
<code>reps</code>	number of simulated point patterns to be generated when computing the envelopes. The default number is 50, a larger number of replications will produce a smoother band, although it takes more time.
<code>conf</code>	the confidence level required for the construction of the envelope. The default is to find 95% confidence envelopes.
<code>type</code>	a character string indicating the type of residuals that should be used in the construction of the envelope. The available options are randomized quantile ("quantile"), standardized ("standard") and studentized ("student") residuals. Standardized and studentized residuals are only available for objects of class "lm" and "ols"; otherwise, quantile residuals are used.
<code>plot.it</code>	if TRUE it will draw the corresponding plot, if FALSE it will only return the computed values.
<code>...</code>	further arguments passed to or from other methods.

Value

a list containing the following elements:

<code>residuals</code>	a vector with the selected (see <code>type</code> argument) residuals.
<code>envelope</code>	a matrix with two columns corresponding to the values of the lower and upper pointwise confidence envelope.

References

Atkinson, A.C. (1985). *Plots, Transformations and Regression*. Oxford University Press, Oxford.

Osorio, F. (2026). On the mean-shift outlier model for LAD regression. *Journal of Statistical Computation and Simulation*. doi:10.1080/00949655.2026.2716387

Venables, W.N., Ripley, B.D. (1999). *Modern Applied Statistics with S-PLUS, 3rd Ed.* Springer, New York.

Examples

```
# QQ-plot with envelope for linear regression
fm <- ols(stack.loss ~ ., data = stackloss)
z <- envelope(fm, reps = 500)

# QQ-plot with envelope for LAD regression
data(ereturns)
fm <- lad(m.marietta ~ CRSP, data = ereturns)
z <- envelope(fm, reps = 500)

# QQ-plot with envelope for ridge regression
data(portland)
fm <- ridge(y ~ ., data = portland)
z <- envelope(fm, reps = 500)

# QQ-plot with envelope nonlinear regression
data(skeena)
model <- recruits ~ b1 * spawners * exp(-b2 * spawners)
fm <- nls(model, data = skeena, start = list(b1 = 3, b2 = 0))
z <- envelope(fm, reps = 500)
```

leverages

Leverages

Description

Computes leverage measures from a fitted model object.

Usage

```
leverages(model, ...)
## S3 method for class 'lm'
leverages(model, infl = lm.influence(model, do.coef = FALSE), ...)
## S3 method for class 'nls'
leverages(model, type = "tangent", ...)
## S3 method for class 'lad'
leverages(model, ...)
## S3 method for class 'ols'
leverages(model, ...)
## S3 method for class 'ridge'
leverages(model, ...)

## S3 method for class 'lad'
hatvalues(model, ...)
## S3 method for class 'ols'
hatvalues(model, ...)
## S3 method for class 'ridge'
hatvalues(model, ...)
```

Arguments

model	an R object, returned by <code>lm</code> , <code>nls</code> , <code>lad</code> , <code>ols</code> or <code>ridge</code> .
infl	influence structure as returned by <code>lm.influence</code> .
type	only required for 'nls' objects, options available are "tangent" and "jacobian" to obtain the tangent plane leverage (Ross, 1987) and Jacobian leverage matrix (St. Laurent and Cook, 1992, 1993), respectively.
...	further arguments passed to or from other methods.

Value

A vector containing the diagonal of the prediction (or 'hat') matrix.

For linear regression (i.e., for "lm" or "ols" objects) the prediction matrix assumes the form

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T,$$

in which case, $h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i$ for $i = 1, \dots, n$. Whereas for ridge regression, the prediction matrix is given by

$$\mathbf{H}(\lambda) = \mathbf{X}(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T,$$

where λ represents the ridge parameter. Thus, the diagonal elements of $\mathbf{H}(\lambda)$, are $h_{ii}(\lambda) = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{x}_i$, $i = 1, \dots, n$.

In nonlinear regression, the tangent plane leverage matrix is given by

$$\hat{\mathbf{H}} = \hat{\mathbf{F}}(\hat{\mathbf{F}}^T \hat{\mathbf{F}})^{-1} \hat{\mathbf{F}}^T,$$

where $\mathbf{F} = \mathbf{F}(\boldsymbol{\beta})$ is the $n \times p$ local model matrix with i th row $\partial f_i(\boldsymbol{\beta}) / \partial \boldsymbol{\beta}$ and $\hat{\mathbf{F}} = \mathbf{F}(\hat{\boldsymbol{\beta}})$, whereas the Jacobian leverage matrix adopts the form:

$$\hat{\mathbf{J}} = \hat{\mathbf{F}}(\hat{\mathbf{F}}^T \hat{\mathbf{F}} - [\hat{\mathbf{r}}^T][\hat{\mathbf{W}}])^{-1} \hat{\mathbf{F}}^T,$$

where $\hat{\mathbf{r}} = \mathbf{Y} - \mathbf{f}(\hat{\boldsymbol{\beta}})$ is the vector of residuals, $\hat{\mathbf{W}}$ is the $n \times p \times p$ three-dimensional array constructed by n Hessian matrices of order $p \times p$ of $\mathbf{f}(\boldsymbol{\beta})$ and $[\cdot][\cdot]$ denotes array multiplication as defined by Bates and Watts (1980).

For LAD regression, this function computes the leverage constants for individual observations introduced by Flores (2015). These constants provide a measure of the relative weight of an observation in the design matrix for LAD regression. Let $\mathbf{X}$ be an $n \times p$ model matrix and let $\mathbf{x}_i^T$ denote the i th row of $\mathbf{X}$. The leverage constant for observation i is defined by

$$\gamma_i = \max_{\substack{\mathbf{g} \in \mathbb{R}^p \\ \|\mathbf{g}\|_2=1}} \frac{|\mathbf{x}_i^T \mathbf{g}|}{\sum_{j=1}^n |\mathbf{x}_j^T \mathbf{g}|}, \quad i = 1, \dots, n.$$

The leverage constants measure the relative weight of individual observations in the design configuration. They correspond to the case $k = 1$ of the leverage constants $c_k(\mathbf{X})$ defined by Flores (2015). In particular,

$$c_1(\mathbf{X}) = 1 - \max_{1 \leq i \leq n} \gamma_i.$$

By homogeneity, the computation of γ_i can equivalently be expressed as the optimization problem

$$\gamma_i = \max_{\mathbf{g} \in \mathbb{R}^p} \{ \mathbf{x}_i^T \mathbf{g} : \|\mathbf{X} \mathbf{g}\|_1 \leq 1 \}.$$

The absolute-value constraint can be formulated as a linear programming problem.

Note

This function never creates the prediction matrix and only obtains its diagonal elements using the QR or singular value decomposition (only for ridge regression) of X or $\tilde{F}$.

Function `hatvalues` only is a wrapper for function `leverages`.

Warning: Starting from version 4.6.0 of R, the function `hatvalues.nls()` will be included in the **stats** package; therefore, it has been removed from **india**. Please use function `leverages.nls()` instead.

For LAD regression, the function `leverages.lad` solves one linear programming problem for each observation. The feasible region is the same for all observations, while the objective function changes with i . The implementation therefore uses the optimal basis obtained for one observation as a warm start for the optimization corresponding to the next observation. The resulting values satisfy, up to numerical precision, $0 \leq \gamma_i \leq 1$. A large value of γ_i indicates that observation i has large leverage according to the definition of Flores (2015).

References

- Bates, D.W., Watts, D.G. (1980). Relative curvature measures of nonlinearity (with discussion). *Journal of the Royal Statistical Society, Series B* **42**, 1-25. doi:10.1111/j.25176161.1980.tb01094.x
- Chatterjee, S., Hadi, A.S. (1988). *Sensitivity Analysis in Linear Regression*. Wiley, New York.
- Cook, R.D., Weisberg, S. (1982). *Residuals and Influence in Regression*. Chapman and Hall, London.
- Flores, S. (2015). Sharp non-asymptotic performance bounds for ℓ_1 and Huber robust regression estimators. *TEST* **24**, 796-812. doi:10.1007/s1174901504355
- Ross, W.H. (1987). The geometry of case deletion and the assessment of influence in nonlinear regression. *The Canadian Journal of Statistics* **15**, 91-103. doi:10.2307/3315198
- St. Laurent, R.T., Cook, R.D. (1992). Leverage and superleverage in nonlinear regression. *Journal of the American Statistical Association* **87**, 985-990. doi:10.1080/01621459.1992.10476253
- St. Laurent, R.T., Cook, R.D. (1993). Leverage, local influence and curvature in nonlinear regression. *Biometrika* **80**, 99-106. doi:10.1093/biomet/80.1.99
- Walker, E., Birch, J.B. (1988). Influence measures in ridge regression. *Technometrics* **30**, 221-227. doi:10.1080/00401706.1988.10488370

Examples

```
# Leverages for linear regression
fm <- ols(stack.loss ~ ., data = stackloss)
levs <- leverages(fm)
cutoff <- 2 * mean(levs)
plot(levs, ylab = "Leverages", ylim = c(0,0.45))
abline(h = cutoff, lty = 2, lwd = 2, col = "red")
text(17, levs[17], label = as.character(17), pos = 3)

# Leverage constants for LAD regression
# first panel of Figure 4 from Flores (2015)
fm <- lad(stack.loss ~ ., data = stackloss, y = TRUE)
levs <- leverages(fm)
```

```

obs <- c(1,2,3,4,17,21)
plot(levs, fm$y, xlab = "Leverages", ylab = "stackloss", xlim = c(0.06,0.2), ylim = c(5,45))
text(levs[obs], fm$y[obs], label = as.character(obs), pos = 3)

# first panel of Figure 2 from Flores (2015)
data(aircraft)
fm <- lad(cost ~ ., data = aircraft, y = TRUE)
levs <- leverages(fm)
obs <- c(14,22)
plot(levs, fm$y, xlab = "Leverages", ylab = "cost", xlim = c(0.05,0.47), ylim = c(0, 115))
text(levs[obs], fm$y[obs], label = as.character(obs), pos = 3)
plot(levs, ylab = "Leverages", ylim = c(0.06,0.48))
text(14, levs[14], label = as.character(14), pos = 3)

# Leverages for ridge regression
data(portland)
fm <- ridge(y ~ ., data = portland)
levs <- leverages(fm)
cutoff <- 2 * mean(levs)
plot(levs, ylab = "Leverages", ylim = c(0,0.7))
abline(h = cutoff, lty = 2, lwd = 2, col = "red")
text(10, levs[10], label = as.character(10), pos = 3)

# Leverages for nonlinear regression
data(skeena)
model <- recruits ~ b1 * spawners * exp(-b2 * spawners)
fm <- nls(model, data = skeena, start = list(b1 = 3, b2 = 0))
lev0 <- leverages(fm, type = "tangent")
plot(lev0, ylab = "Tangent plane leverages", ylim = c(0,0.25))
obs <- c(1,9)
text(obs, lev0[obs], label = as.character(obs), pos = 3)
lev1 <- leverages(fm, type = "jacobian")
plot(lev1, ylab = "Jacobian leverages", ylim = c(0,0.25))
obs <- c(1,9)
text(obs, lev1[obs], label = as.character(obs), pos = 3)
# in this example both leverages (tangent and jacobian)
# are quite similar
plot(lev0, lev1, xlab = "Tangent plane leverages",
     ylab = "Jacobian leverages")
abline(c(0,1), lty = 2, lwd = 2, col = "red")

```

Description

Compute the likelihood displacement influence measure based on leave-one-out cases deletion for linear models, lad and ridge regression.

Usage

```
logLik.displacement(model, ...)
## S3 method for class 'lm'
logLik.displacement(model, pars = "full", ...)
## S3 method for class 'ols'
logLik.displacement(model, pars = "full", ...)
## S3 method for class 'nls'
logLik.displacement(model, ...)
## S3 method for class 'lad'
logLik.displacement(model, method = "quasi", pars = "full", ...)
## S3 method for class 'ridge'
logLik.displacement(model, pars = "full", ...)
```

Arguments

model	an R object, returned by <code>lm</code> , <code>nls</code> , <code>ols</code> , <code>lad</code> or <code>ridge</code> .
pars	should be considered the whole vector of parameters (<code>pars = "full"</code>), or only the vector of coefficients (<code>pars = "coef"</code>). This option is not used for <code>nls</code> objects.
method	only required for 'lad' objects, options available are "quasi" and "BR" to obtain the likelihood displacement based on Sun and Wei (2004), and Elian et al. (2000) approaches, respectively.
...	further arguments passed to or from other methods.

Value

A vector whose i th element contains the distance between the likelihood functions,

$$LD_i(\beta, \sigma^2) = 2\{l(\hat{\beta}, \hat{\sigma}^2) - l(\hat{\beta}_{(i)}, \hat{\sigma}_{(i)}^2)\},$$

for `pars = "full"`, where $\hat{\beta}_{(i)}$ and $\hat{\sigma}_{(i)}^2$ denote the estimates of β and σ^2 when the i th observation is removed from the dataset. If we are interested only in β (i.e. `pars = "coef"`) the likelihood displacement becomes

$$LD_i(\beta|\sigma^2) = 2\{l(\hat{\beta}, \hat{\sigma}^2) - \max_{\sigma^2} l(\hat{\beta}_{(i)}, \hat{\sigma}^2)\}.$$

References

- Cook, R.D., Weisberg, S. (1982). *Residuals and Influence in Regression*. Chapman and Hall, London.
- Cook, R.D., Pena, D., Weisberg, S. (1988). The likelihood displacement: A unifying principle for influence measures. *Communications in Statistics - Theory and Methods* **17**, 623-640. doi:10.1080/03610928808829645
- Elian, S.N., Andre, C.D.S., Narula, S.C. (2000). Influence measure for the L1 regression. *Communications in Statistics - Theory and Methods* **29**, 837-849. doi:10.1080/03610920008832518
- Ogueda, A., Osorio, F. (2025). Influence diagnostics for ridge regression using the Kullback-Leibler divergence. *Statistical Papers* **66**, 85. doi:10.1007/s00362025017011

Ross, W.H. (1987). The geometry of case deletion and the assessment of influence in nonlinear regression. *The Canadian Journal of Statistics* **15**, 91-103. doi:10.2307/3315198

Sun, R.B., Wei, B.C. (2004). On influence assessment for LAD regression. *Statistics & Probability Letters* **67**, 97-110. doi:10.1016/j.spl.2003.08.018

Examples

```
# Likelihood displacement for linear regression
fm <- ols(stack.loss ~ ., data = stackloss)
LD <- logLik.displacement(fm)
plot(LD, ylab = "Likelihood displacement", ylim = c(0,9))
text(21, LD[21], label = as.character(21), pos = 3)

# Likelihood displacement for LAD regression
fm <- lad(stack.loss ~ ., data = stackloss)
LD <- logLik.displacement(fm)
plot(LD, ylab = "Likelihood displacement", ylim = c(0,1.5))
text(17, LD[17], label = as.character(17), pos = 3)

# Likelihood displacement for ridge regression
data(portland)
fm <- ridge(y ~ ., data = portland)
LD <- logLik.displacement(fm)
plot(LD, ylab = "Likelihood displacement", ylim = c(0,4))
text(8, LD[8], label = as.character(8), pos = 3)

# Likelihood displacement for nonlinear regression
data(skeena)
model <- recruits ~ b1 * spawners * exp(-b2 * spawners)
fm <- nls(model, data = skeena, start = list(b1 = 3, b2 = 0))
LD <- logLik.displacement(fm)
plot(LD, ylab = "Likelihood displacement", ylim = c(0,0.7))
obs <- c(5, 6, 9, 19, 25)
text(obs, LD[obs], label = as.character(obs), pos = 3)
```

portland	<i>Portland cement dataset</i>
----------	--------------------------------

Description

This dataset comes from an experimental investigation of the heat evolved during the setting and hardening of Portland cements of varied composition and the dependence of this heat on the percentages of four compounds in the clinkers from which the cement was produced.

Usage

```
data(portland)
```

Format

- A data frame with 13 observations on the following 5 variables.
- y** The heat evolved after 180 days of curing, measured in calories per gram of cement.
- x1** Tricalcium aluminate.
- x2** Tricalcium silicate.
- x3** Tetracalcium aluminoferrite.
- x4** β -dicalcium silicate.

Source

Kaciranlar, S., Sakallioğlu, S., Akdeniz, F., Styán, G.P.H., Werner, H.J. (1999). A new biased estimator in linear regression and a detailed analysis of the widely-analysed dataset on Portland cement. *Sankhya, Series B* **61**, 443-459.

relative.condition	<i>Relative change in the condition number</i>
--------------------	--

Description

Compute the relative condition index to identify collinearity-influential points in linear models.

Usage

relative.condition(x)

Arguments

x the model matrix $\mathbf{X}$.

Value

To assess the influence of the i th row of $\mathbf{X}$ on the condition index of $\mathbf{X}$, Hadi (1988) proposed the relative change,

$$\delta_i = \frac{\kappa_{(i)} - \kappa}{\kappa},$$

for $i = 1, \dots, n$, where $\kappa = \kappa(\mathbf{X})$ and $\kappa_{(i)} = \kappa(\mathbf{X}_{(i)})$ denote the (scaled) condition index for $\mathbf{X}$ and $\mathbf{X}_{(i)}$, respectively.

References

Chatterjee, S., Hadi, A.S. (1988). *Sensitivity Analysis in Linear Regression*. Wiley, New York.

Hadi, A.S. (1988). Diagnosing collinearity-influential observations. *Computational Statistics & Data Analysis* **7**, 143-159. doi:10.1016/01679473(88)900898.

Examples

```
data(portland)
fm <- ridge(y ~ ., data = portland, x = TRUE)
x <- fm$x
rel <- relative.condition(x)
plot(rel, ylab = "Relative condition number", ylim = c(-0.1,0.4))
abline(h = 0, lty = 2, lwd = 2, col = "red")
text(3, rel[3], label = as.character(3), pos = 3)
```

rquantile

*Randomized quantile residuals***Description**

Compute randomized quantile residuals from a fitted model object.

Usage

```
rquantile(model, ...)
## S3 method for class 'lm'
rquantile(model, ...)
## S3 method for class 'lad'
rquantile(model, ...)
## S3 method for class 'ols'
rquantile(model, ...)
## S3 method for class 'nls'
rquantile(model, ...)
## S3 method for class 'ridge'
rquantile(model, ...)
```

Arguments

`model` an R object, returned by `lm`, `lad`, `ols`, `nls` or `ridge`.
`...` further arguments passed to or from other methods.

Value

a vector containing standard normal deviates representing standardized residuals. This kind of residuals are exactly normal.

References

Dunn, P.K., Smyth, G.K. (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics* **5**, 236-244. doi:[10.1080/10618600.1996.10474708](https://doi.org/10.1080/10618600.1996.10474708)
Osorio, F. (2026). On the mean-shift outlier model for LAD regression. *Journal of Statistical Computation and Simulation*. doi:[10.1080/00949655.2026.2716387](https://doi.org/10.1080/00949655.2026.2716387)

Examples

```
# quantile residuals for linear regression
fm <- ols(stack.loss ~ ., data = stackloss)
res <- rquantile(fm)
plot(res, ylim = c(-3,3), ylab = "quantile residuals")
abline(h = 0, lwd = 2, col = "gray75")
abline(h = c(-2,2), lwd = 2, lty = 2, col = "red")
text(21, res[21], as.character(21), pos = 1)

# quantile residuals for LAD regression
data(ereturns)
fm <- lad(m.marietta ~ CRSP, data = ereturns)
res <- rquantile(fm)
plot(res, ylim = c(-2,4.5), ylab = "quantile residuals")
abline(h = 0, lwd = 2, col = "gray75")
abline(h = c(-2,2), lwd = 2, lty = 2, col = "red")
obs <- c(8,15,34)
text(obs, res[obs], as.character(obs), pos = 3)

# quantile residuals for ridge regression
data(portland)
fm <- ridge(y ~ ., data = portland)
res <- rquantile(fm)
plot(res, ylim = c(-2,2), ylab = "quantile residuals")
abline(h = 0, lwd = 2, col = "gray75")

# quantile residuals for nonlinear regression
data(skeena)
model <- recruits ~ b1 * spawners * exp(-b2 * spawners)
fm <- nls(model, data = skeena, start = list(b1 = 3, b2 = 0))
res <- rquantile(fm)
plot(res, ylim = c(-3,3), ylab = "quantile residuals")
abline(h = 0, lwd = 2, col = "gray75")
abline(h = c(-2,2), lwd = 2, lty = 2, col = "red")
text(5, res[5], as.character(5), pos = 3)
```

skeena

Skeena River sockeye salmon data

Description

The data have 28 observations of spawners and recruits (units are thousands of fish) from 1940 until 1967 for the Skeena river sockeye salmon stock.

Usage

```
data(skeena)
```

Format

A data frame with 28 observations on the following 3 variables.

year Years in which the number of spawners and recruits were recorded.

spawners Size of the annual spawning stock.

recruits Production of new catchable-sized fish.

Source

Carroll, R.J., Ruppert, D. (1988). *Transformation and Weighting in Regression*. Chapman and Hall, London.

Index

* datasets

- aircraft, [2](#)
- portland, [11](#)
- skeena, [14](#)

* regression

- cooks.distance, [2](#)
- envelope, [4](#)
- leverages, [6](#)
- logLik.displacement, [9](#)
- relative.condition, [12](#)
- rquantile, [13](#)

aircraft, [2](#)

cooks.distance, [2](#)

envelope, [4](#)

hatvalues, [8](#)

hatvalues.lad (leverages), [6](#)

hatvalues.ols (leverages), [6](#)

hatvalues.ridge (leverages), [6](#)

lad, [5](#), [13](#)

leverages, [6](#)

lm, [5](#), [7](#), [10](#), [13](#)

lm.influence, [7](#)

logLik.displacement, [9](#)

nls, [3](#), [5](#), [7](#), [10](#), [13](#)

ols, [5](#), [13](#)

portland, [11](#)

relative.condition, [12](#)

ridge, [5](#), [13](#)

rquantile, [13](#)

skeena, [14](#)